

# Optimization of a UAV for Wildfire Management

Nathaniel Hargan

The problem is formulated as follows:

$$\min \text{mass}(d_a, t_a, d_s, t_s, r_s, t_f, t_w)$$

subject to:

$$g_j(x) \leq 0; j = 0, 1, 2, \dots, J$$

$$x_i^L \leq x_i \leq x_i^U$$

design variables:

- $d_a$ : arm diameter,
- $t_a$ : arm thickness,
- $d_s$ : strut diameter,
- $t_s$ : strut thickness,
- $r_s$ : strut distance from center,
- $t_f$ : hub flange thickness,
- $t_w$ : hub web thickness

derived variables:

- $d_{tip}$ : Deflection at tip
- $\sigma_{max}$ : Maximum stress from bending
- $f_{amp}$ : Amplitude of displacement at the tip of the uav arm
- $D$ : Damage per mission

design parameters:

- $d_{allowable}$ : Deflection allowable
- $\sigma_{allowable}$ : Stress allowable from bending

- $R_{allowable}$ : Allowable amplitude displacement ratio
- $n_{missions}$ : numbr of missions before damage

Constraints:

Limit on deflection from bending by force required to lift the UAV.

$$g_1(d_a, t_a, d_s, t_s, r_s, t_f, t_w) = d_{tip} \leq d_{allowable}$$

Limit on stress from bending by force required to lift the UAV.

$$g_2(d_a, t_a, d_s, t_s, r_s, t_f, t_w) = \sigma_{max} \leq \sigma_{allowable}$$

Limit on amplitude of displacement given a forcing function based on the propeller speed of the UAV.

$$g_3(d_a, t_a, d_s, t_s, r_s, t_f, t_w) = \frac{f_{amp}}{d_{tip}} \leq R_{allowable}$$

Limit on the damage the UAV during a mission.

$$g_4(d_a, t_a, d_s, t_s, r_s, t_f, t_w) = D * n_{missions} \leq 1$$

Energys